

# A JOURNEY THROUGH INNOVATION AND IMPACT

HOW SEMICONDUCTORS REVOLUTIONIZED THE WORLD

RONALD J. TACKETT, PH.D.  
ASSOCIATE PROFESSOR OF PHYSICS  
KETTERING UNIVERSITY  
[rtackett@kettering.edu](mailto:rtackett@kettering.edu)



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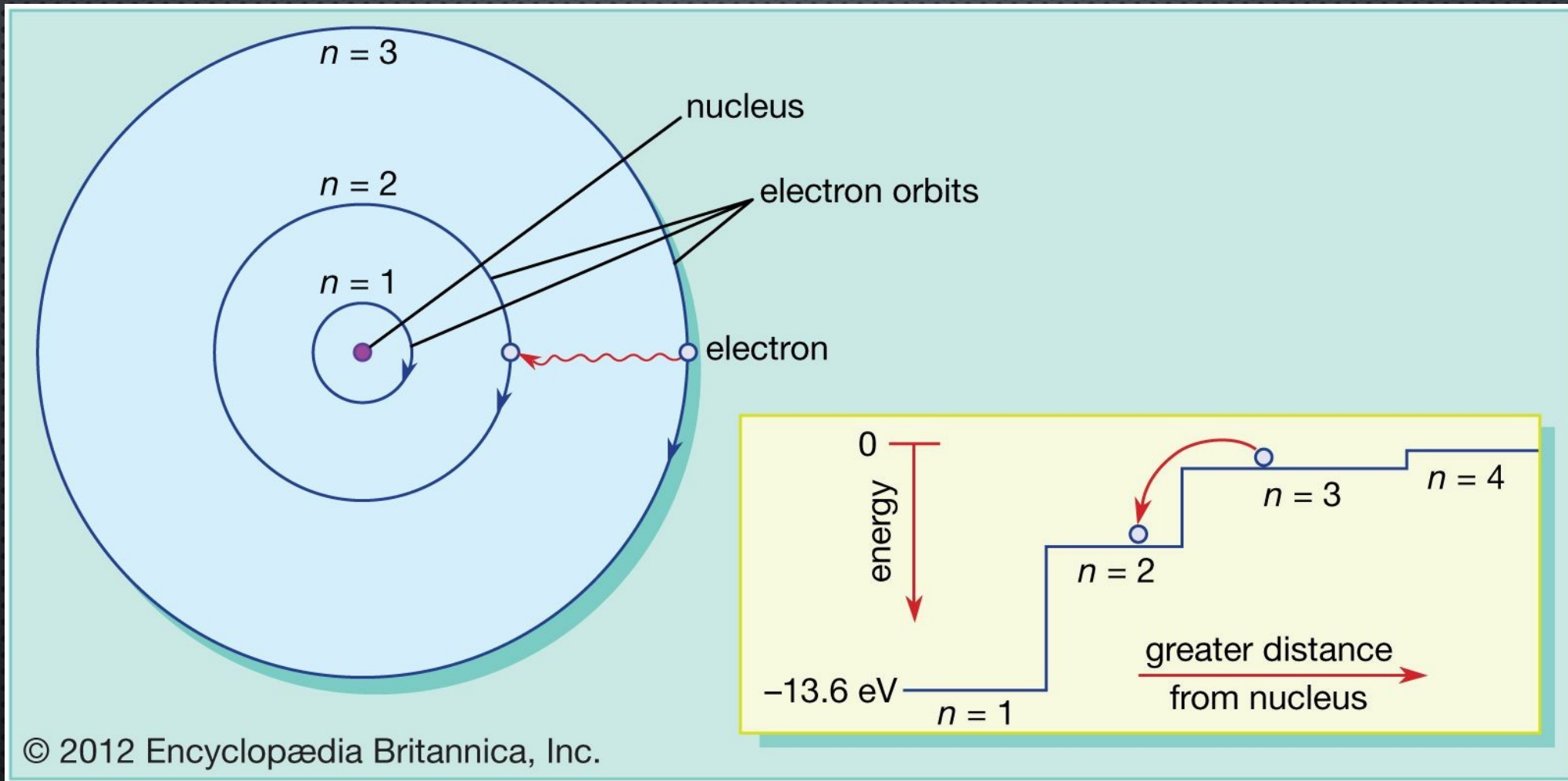


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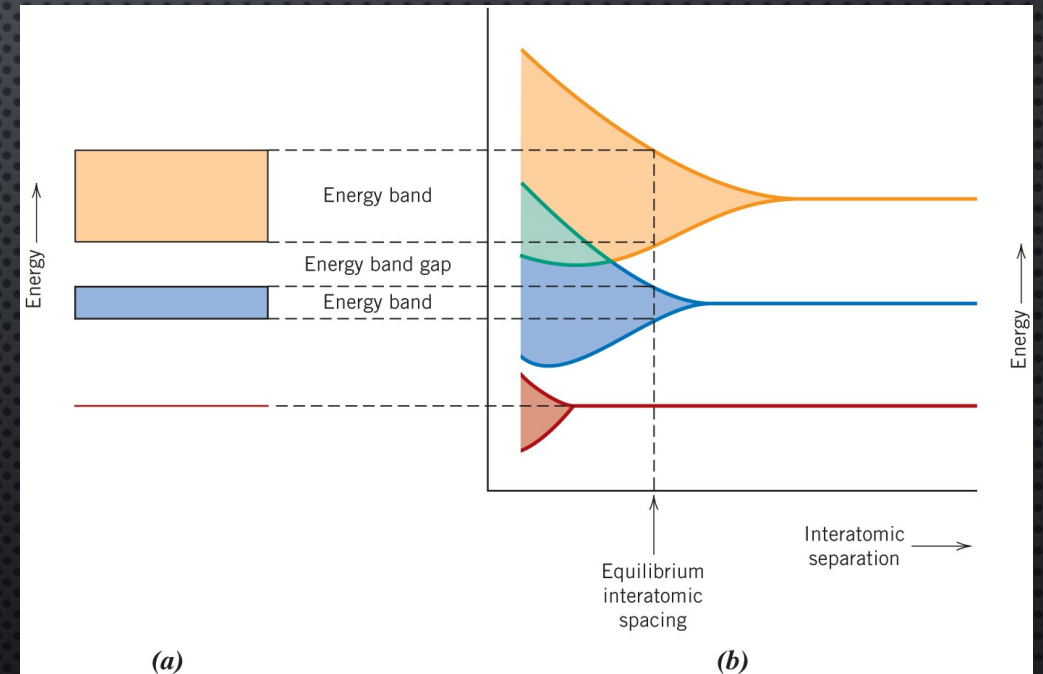
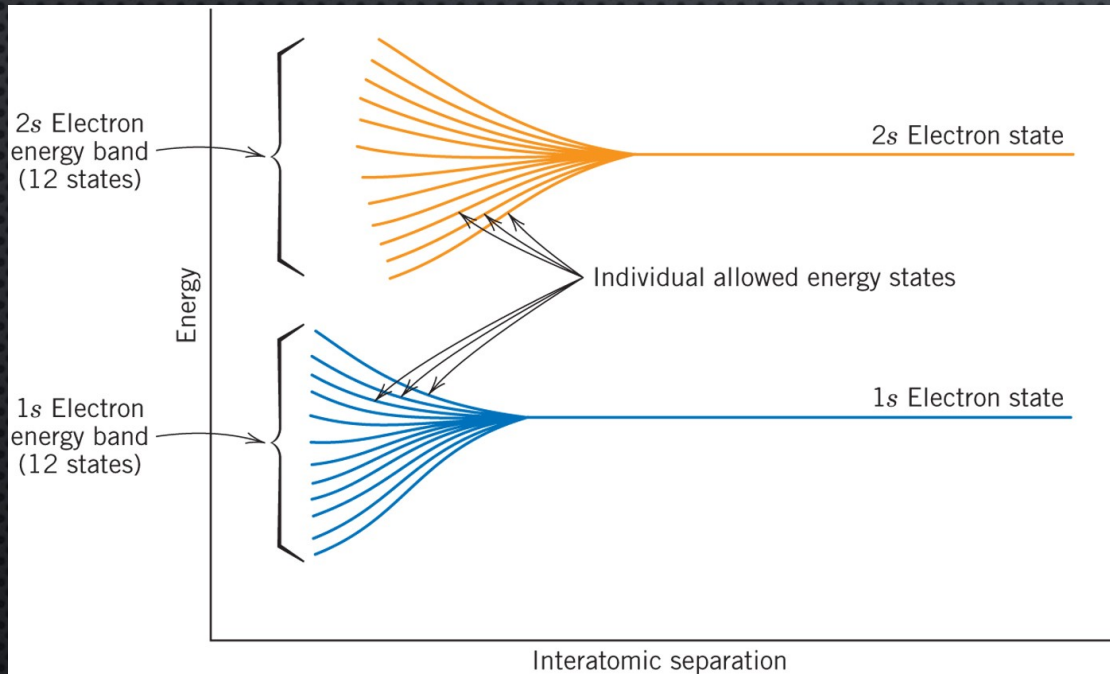
# ~~What is a semiconductor?~~ What is electricity?

Electrons (which most commonly exist in atoms) lie at the heart of electricity



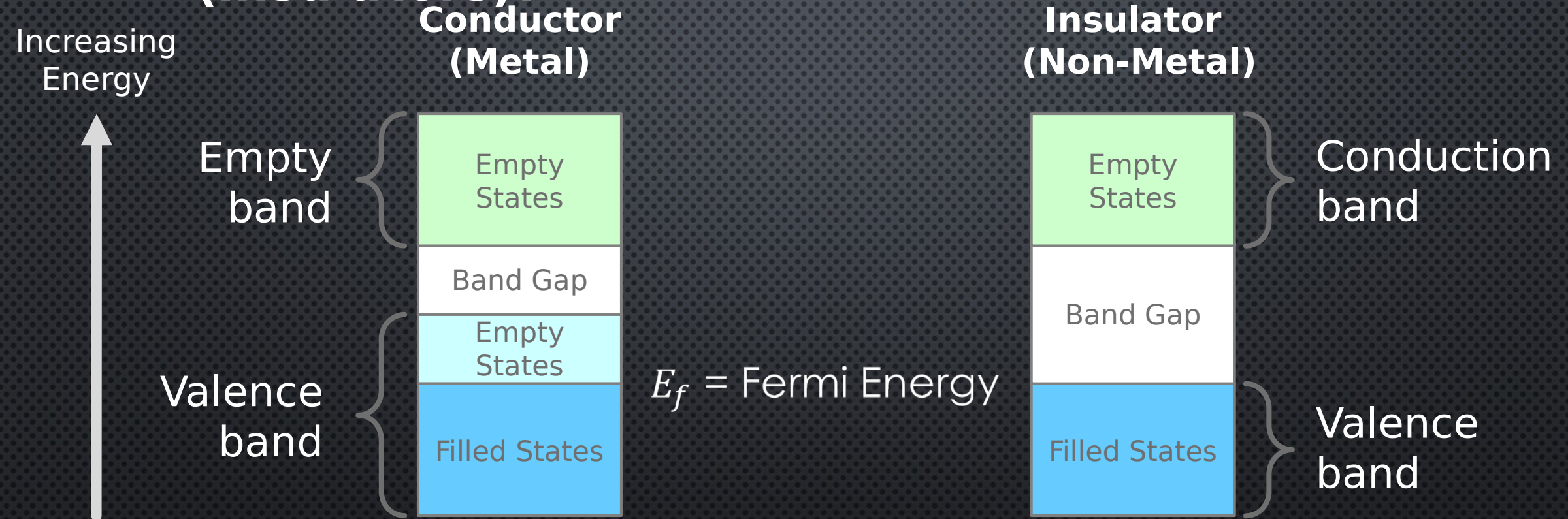
# ~~What is a semiconductor?~~ How do electrons move through solid materials (what we call electricity)?

## What happens when we form bonds between atoms to make a solid material?





# ~~What is a semiconductor?~~ are conductors (metals) and non-conductors (insulators)?

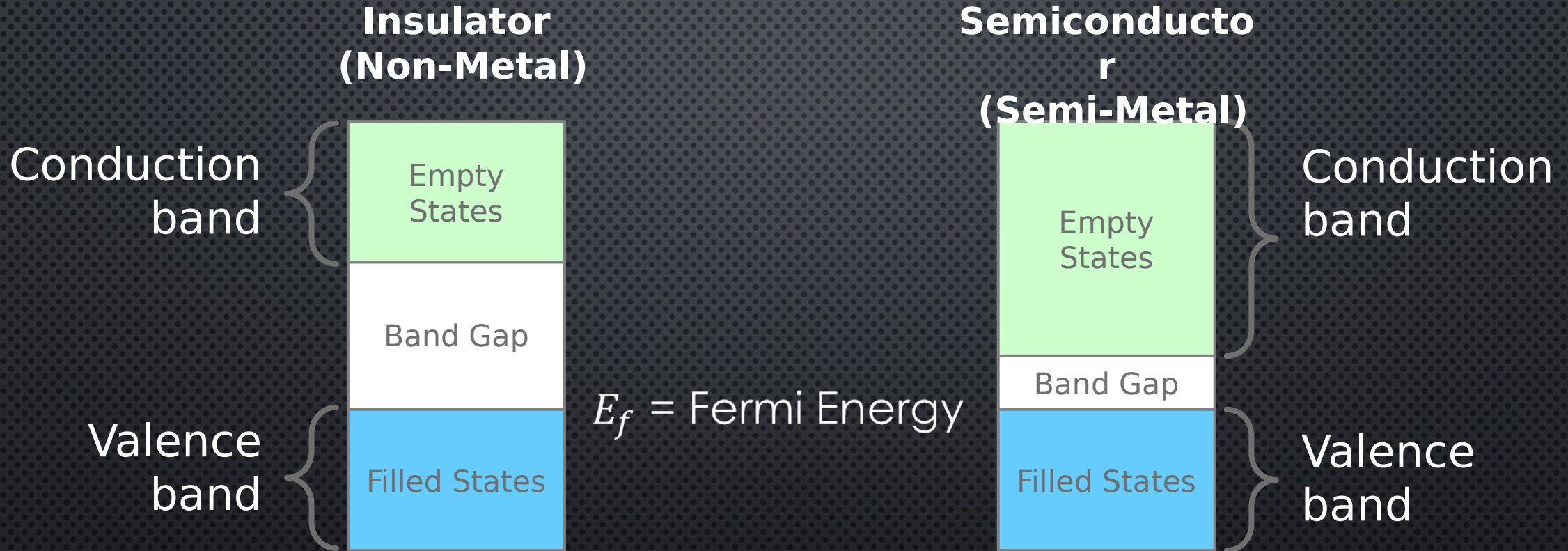


In order to participate in the conduction process, electrons must be free from their atoms, which requires that they have more energy than the Fermi energy.



# What is a semiconductor?

The only difference between an insulator and a semiconductor is the width of the band gap.



In this sense, ***intrinsic*** semiconductors are materials that are (1) not really good conductors (filled valence band separated by a band gap from the conduction band), and (2) not really good insulators (the small band gap leads to higher than usual conductivity).



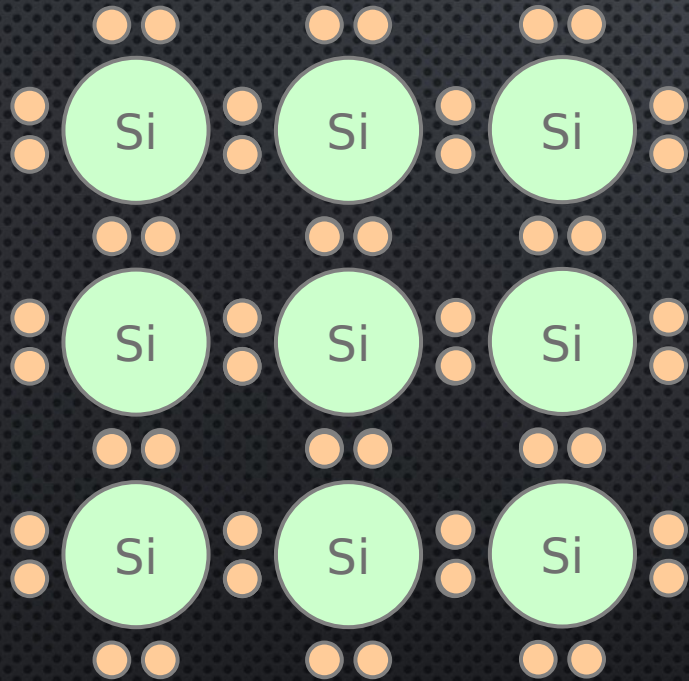
**How did a class of  
materials that isn't  
really good at  
anything  
(electrically)  
completely  
transform the  
electronics industry**



# The answer: performance enhancing drugs (doping)

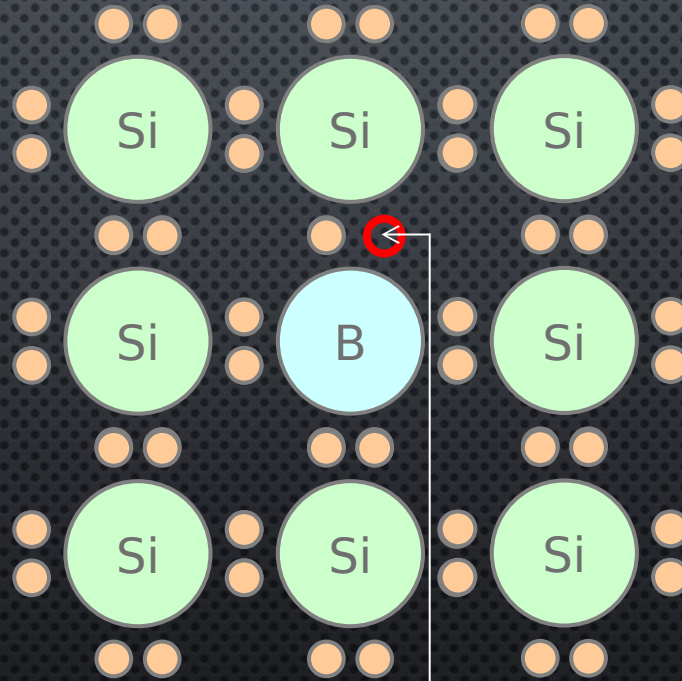
## Intrinsic (Undoped)

pure silicon

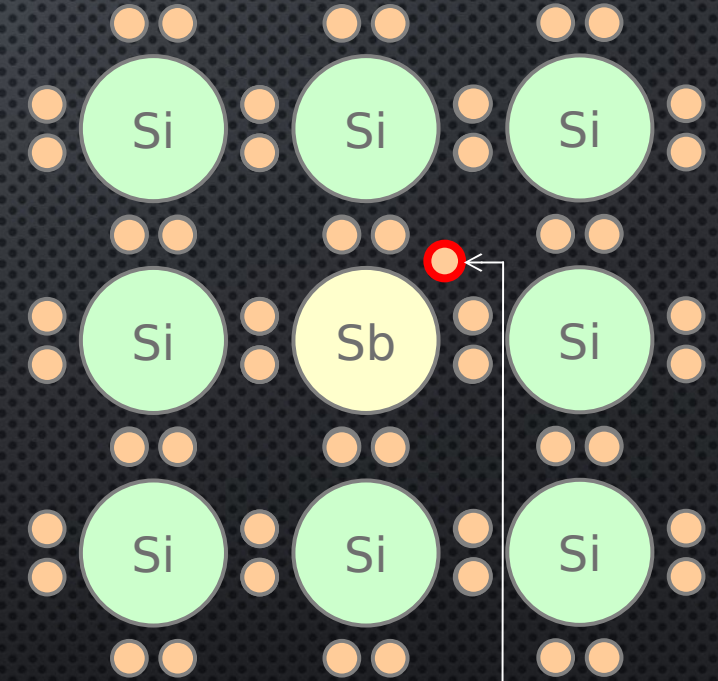


## Extrinsic (Doped)

*p*-type silicon



*n*-type silicon



Doping allows for the precise control of the carrier concentration in these materials



# Periodic Table of the Elements

Kettering  
UNIVERSITY



Doping intrinsic semiconductor materials (which makes extrinsic semiconductor materials) allows us to:

1. Precisely control the electrical conductivity of the material (the ease with which charge can flow through the material)
2. Control the type (sign) of the charge carriers in the material



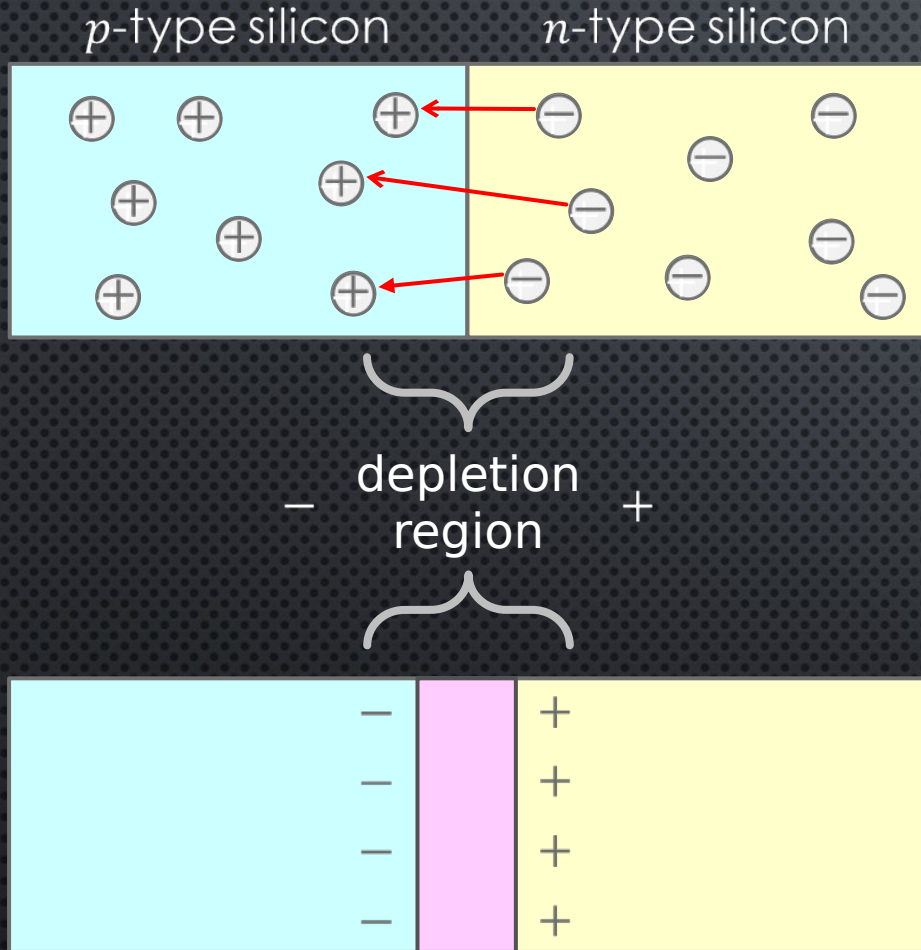
**You may still be thinking:**

**Just being able to  
precisely control the  
conductivity and carrier  
type still doesn't seem  
like something  
REVOLUTIONARY...**

**What's the deal?**



# Where the magic happens: The $p$ - $n$ junction (diode)



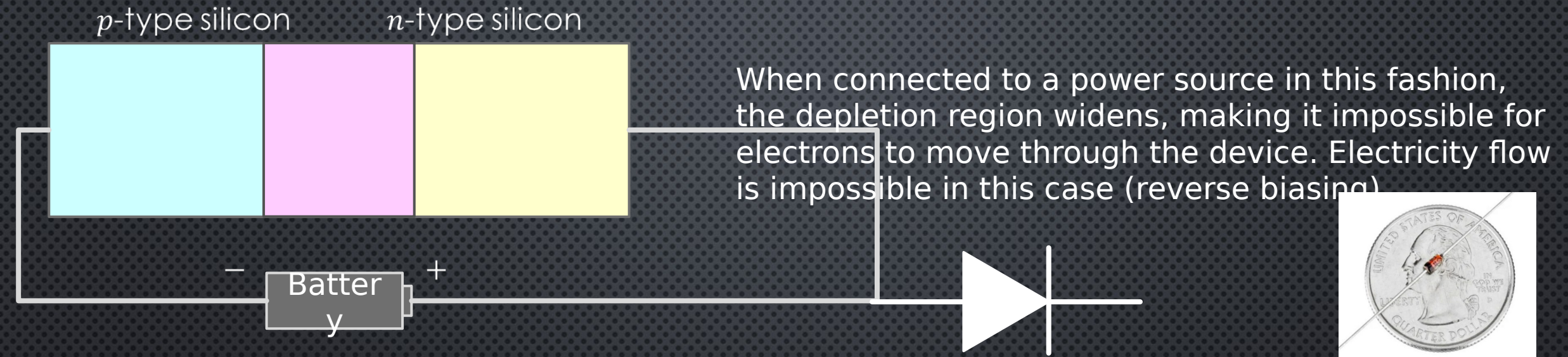
At the  $pn$  junction, the abundant **free electrons** on the  $n$  side tend to migrate to the **holes** that are available on the  $p$  side.

This results in the creation of a **depletion region** and the build up of a **potential barrier** that opposes charge flow through the device

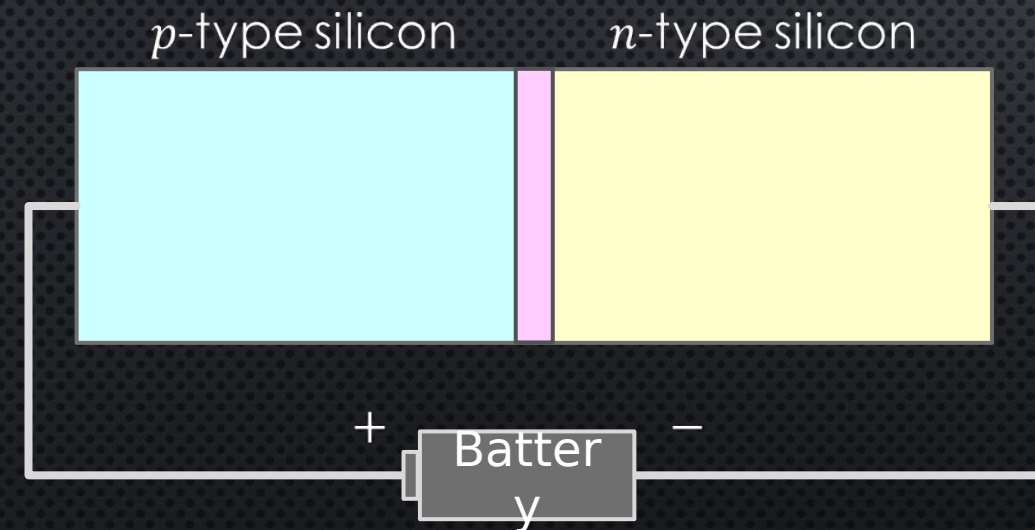
Barrier Potential  $\approx 0.7$  V for silicon-based diodes



# Where the magic happens: The $p$ - $n$ junction (diode)



<https://www.addicore.com/>

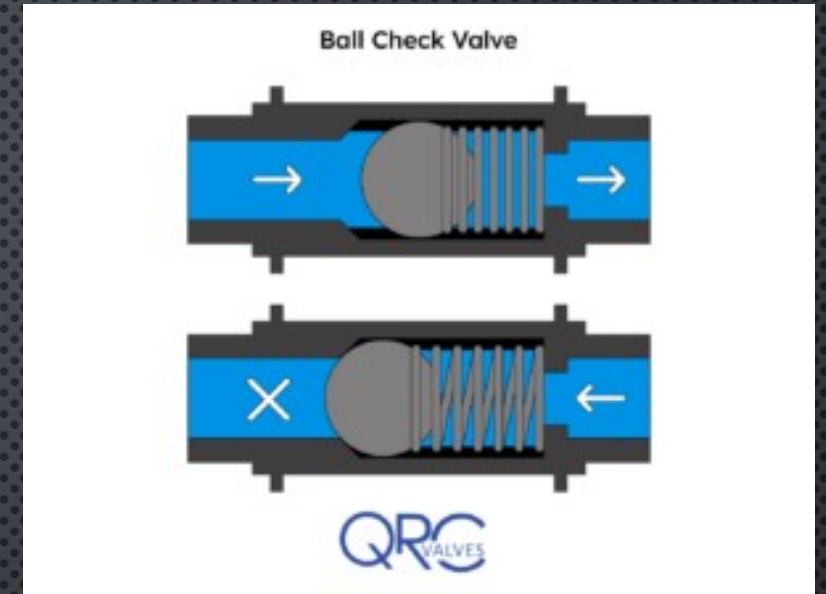
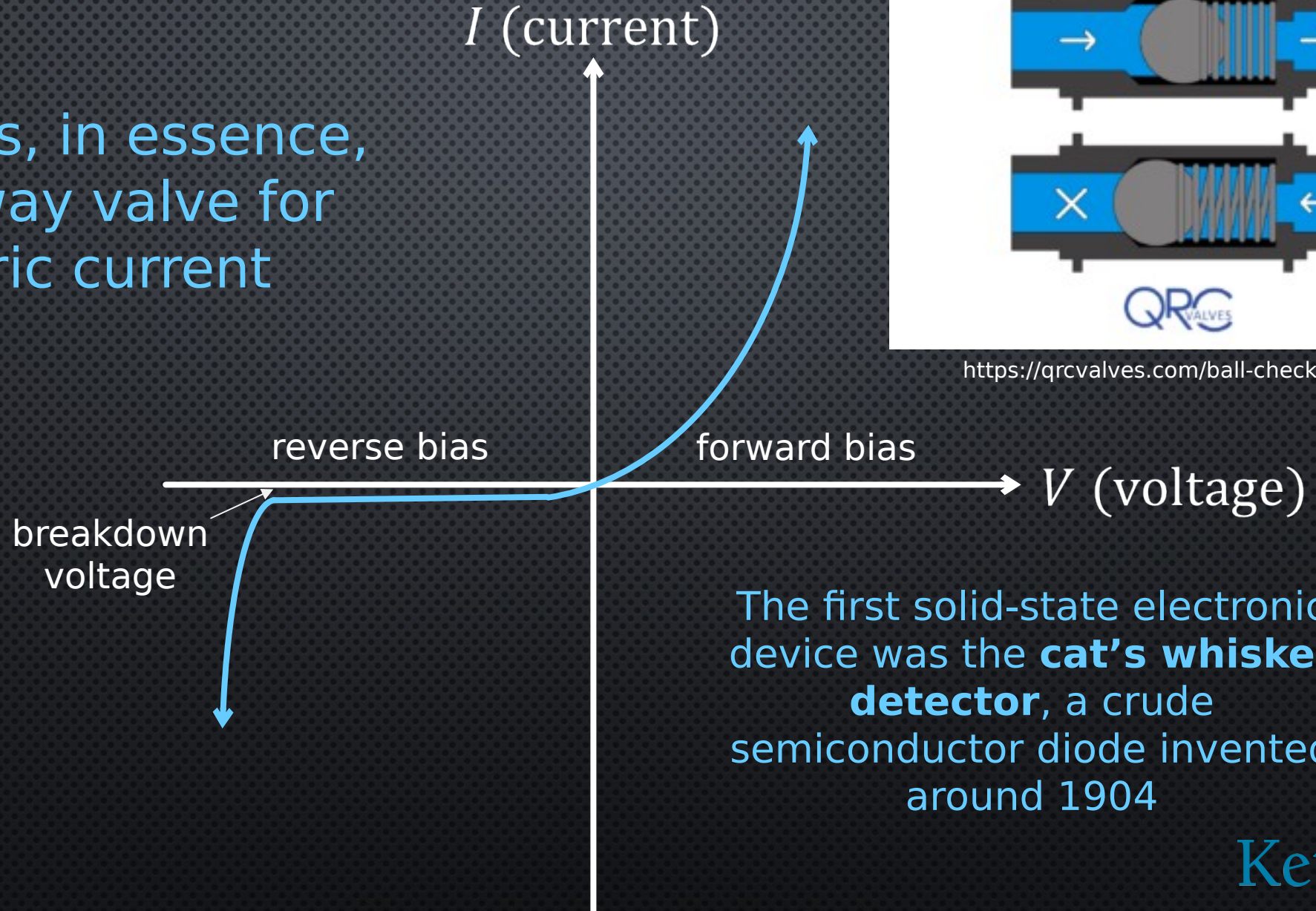


Note the difference in the orientation of the battery in the two cases



# The $I$ $V$ characteristics of a diode

A diode is, in essence,  
a one-way valve for  
electric current

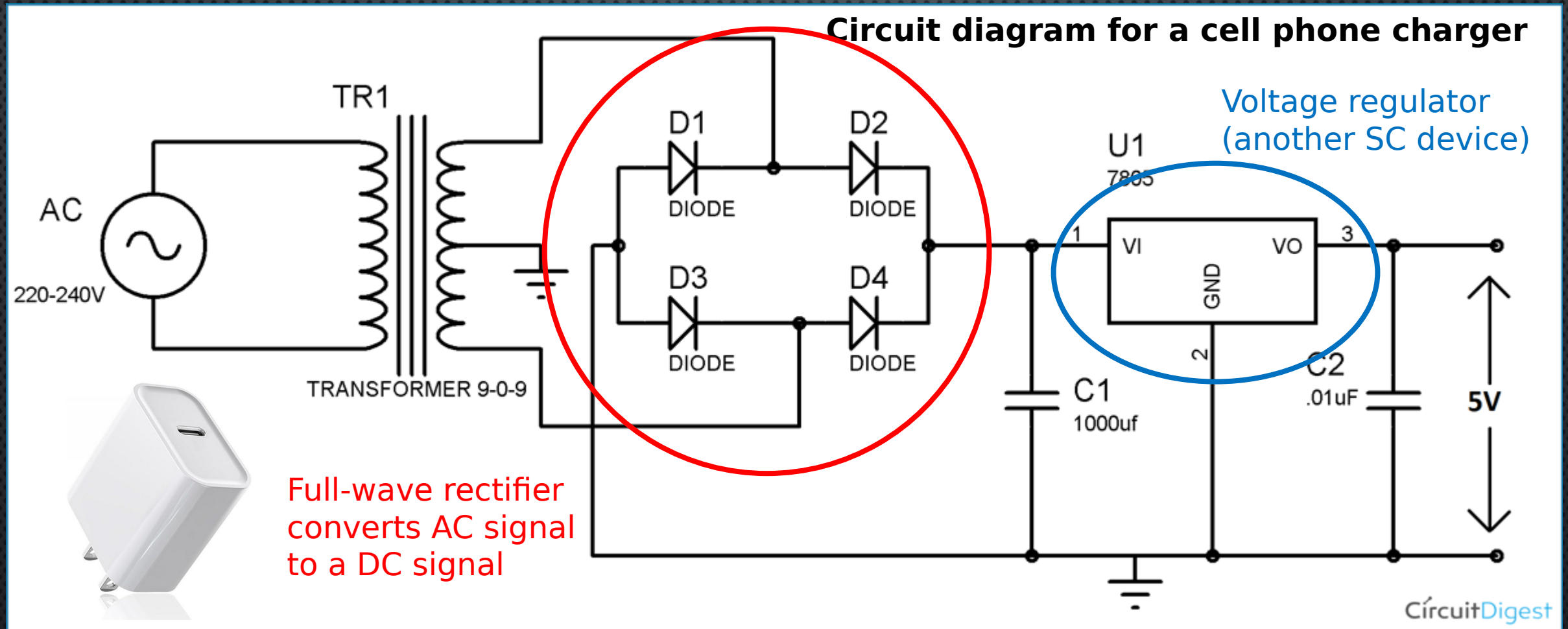


<https://qrcvalves.com/ball-check-valves/>

The first solid-state electronic device was the **cat's whisker detector**, a crude semiconductor diode invented around 1904



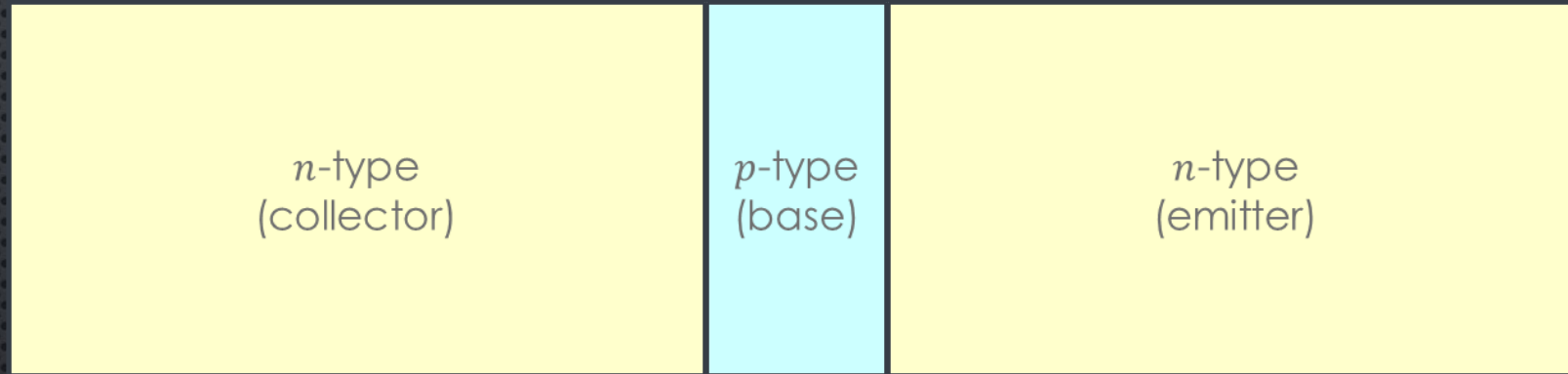
# The *pn*-junction is often called the *rectifying junction*





# The Bipolar Junction Transistor (BJT)

Start with a single-crystal intrinsic semiconductor



<https://www.sparkfun.com/>

We then make the ends heavily-doped (lots of free charge carriers) *n*-type, leaving a gap in the middle

Finally, we make the central region lightly-doped (less charge carriers than the ends) *p*-type

We refer to this as an *npn* bipolar junction transistor  
(It is, essentially, two diodes, back-to-back)





**If you place two diodes  
back-to-back, it would  
seem that current  
would never be able to  
flow through the  
device, right?**

**How is this**

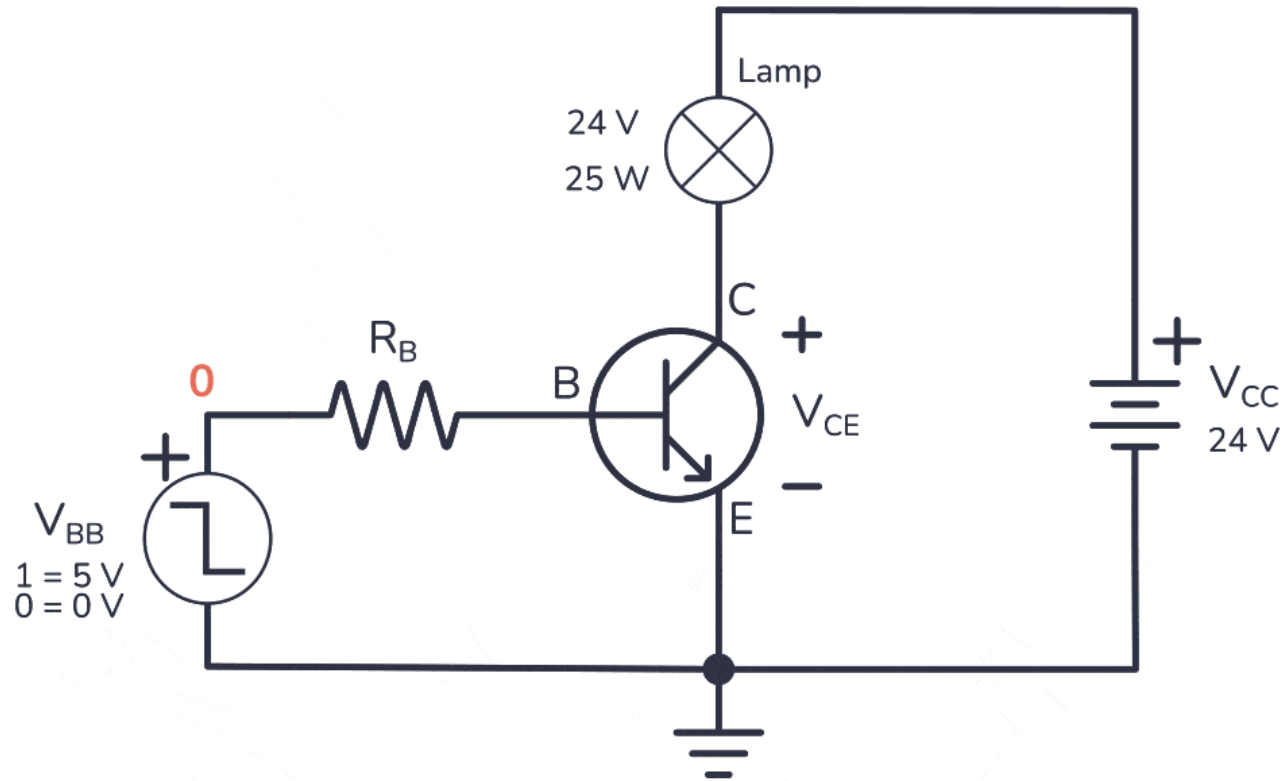


This (six-minute) video begins to explain it (and will give me a break.)





# The BJT as a Switch...



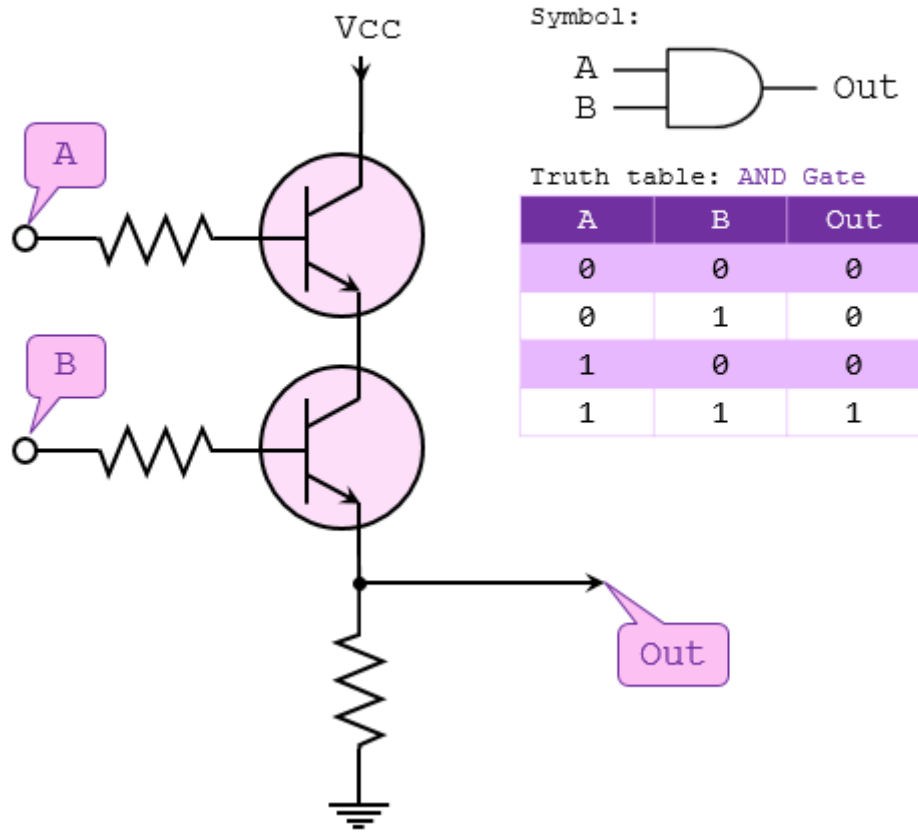
When the input signal is set to 0V, no current flows through  $R_B$ , causing the transistor to act as an open switch, and the lamp does not light

When the input signal is set to 5V, current flows through  $R_B$ , causing the transistor to act as a closed switch, turning on the lamp

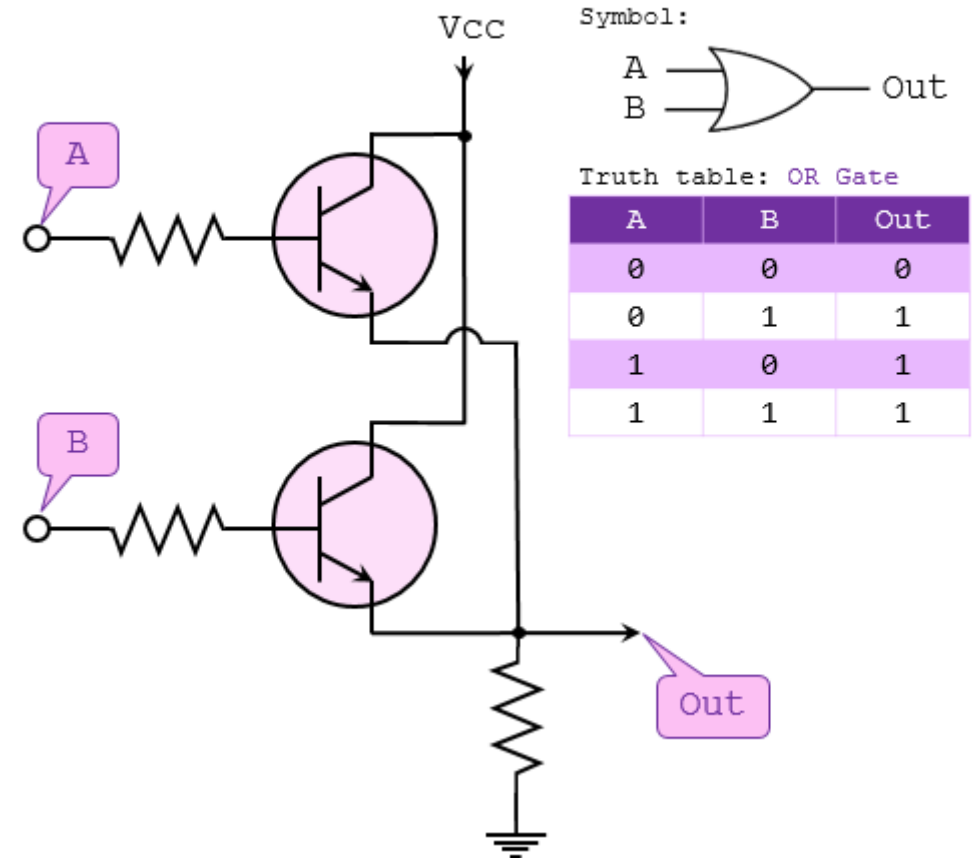


# The BJT as a Logic Gate

## AND Gate



## OR Gate

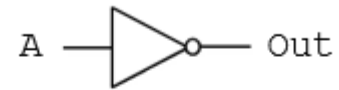




# The BJT as a Logic Gate

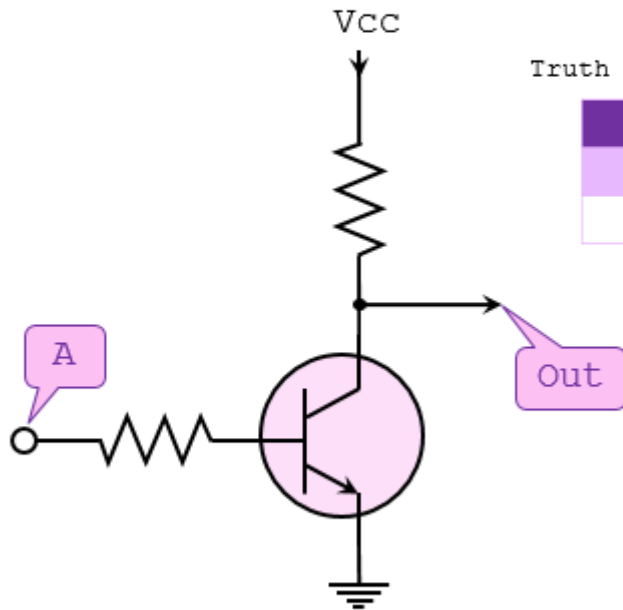
## NOT Gate

Symbol:



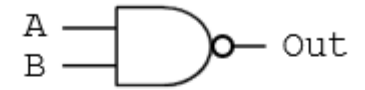
Truth table: NOT Gate

| A | Out |
|---|-----|
| 0 | 1   |
| 1 | 0   |



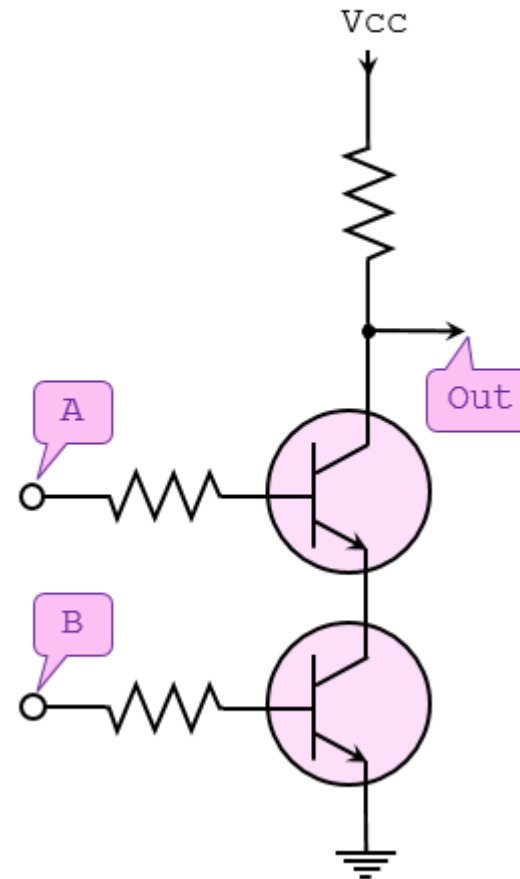
## NAND Gate

Symbol:



Truth table: NAND Gate

| A | B | Out |
|---|---|-----|
| 0 | 0 | 1   |
| 0 | 1 | 1   |
| 1 | 0 | 1   |
| 1 | 1 | 0   |





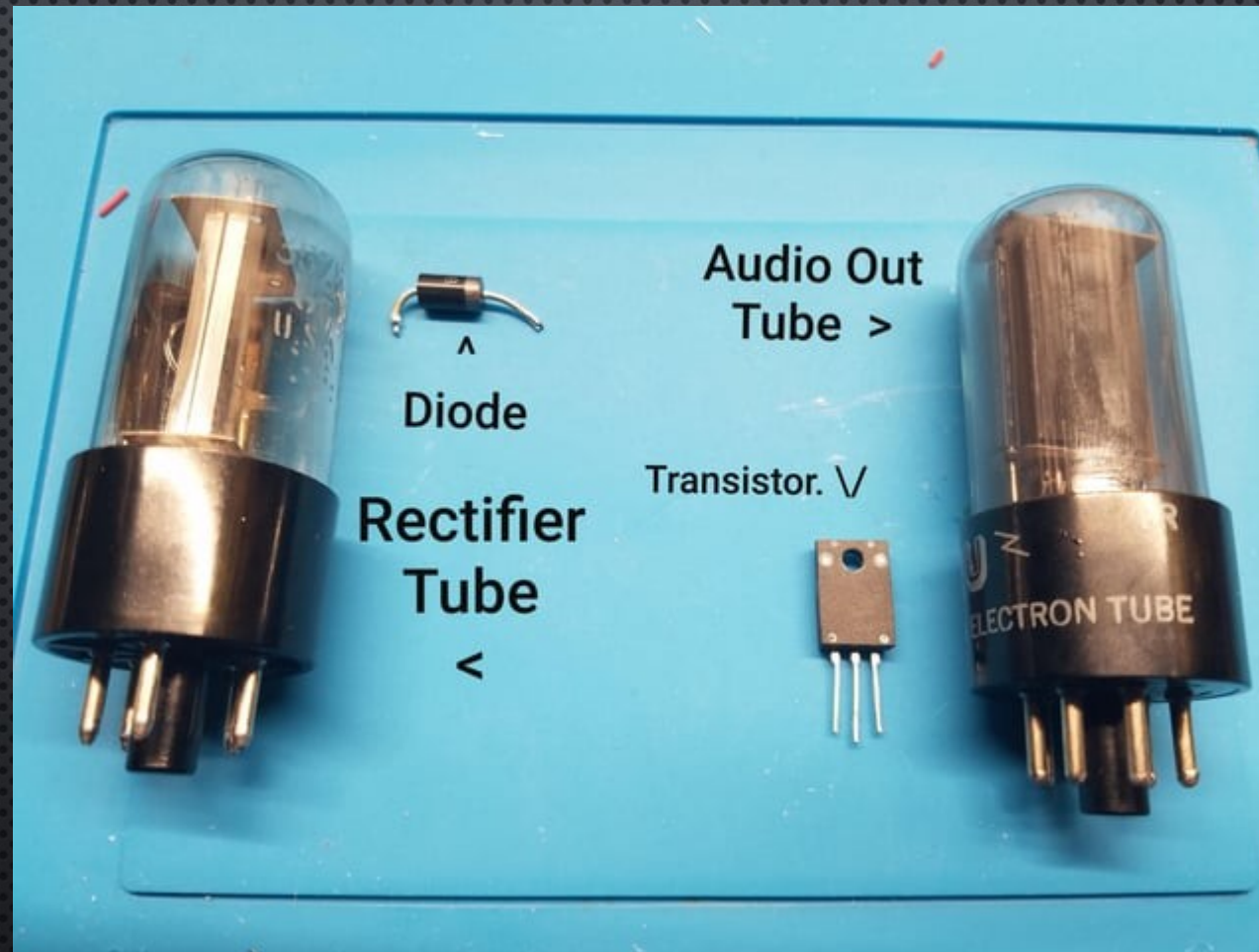
# But how does all of this lead to a revolution?

Especially considering the fact that we already had devices (vacuum-tube technology) that could already do what transistors and diodes do...



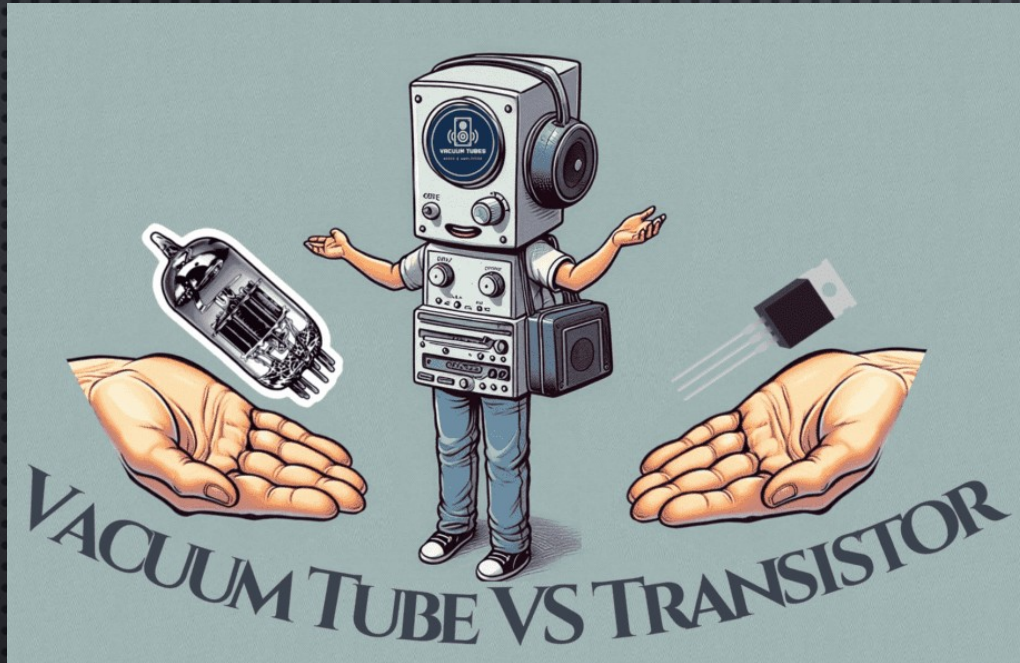
# The answer is size (and power consumption)

...and a little bit of manufacturability





# The debate rages on amongst audiophiles...



## Vacuum Tubes:

**Operation:** Vacuum tubes function through thermionic emission, where electrons are emitted from a heated cathode in a vacuum. This process is more analog and provides a smooth signal transition.

**Sound Quality:** Known for their warm, rich tones and smooth distortion, vacuum tubes are preferred in high-fidelity audio equipment and musical instrument amplifiers. They deliver a more natural sound due to the analog nature of the amplification.

## Transistors:

**Operation:** Transistors work using semiconductor materials to amplify signals. They switch on and off quickly, making them efficient and suitable for various electronic applications.

**Power and Efficiency:** Transistors consume less power and generate less heat compared to vacuum tubes. They are smaller, more durable, and reliable, making them ideal for modern electronic devices.

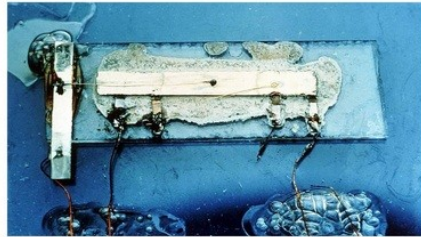
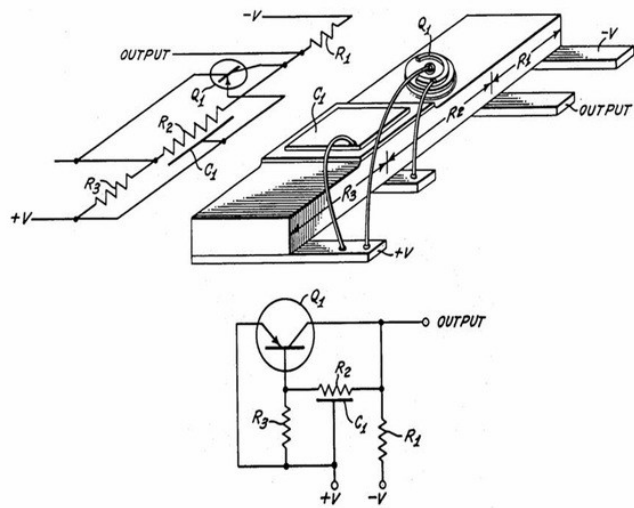


# The Integrated-Circuit (IC) is Born...

## The First (2D) Integrated Circuit

Jack Kilby, Texas Instruments, 1958

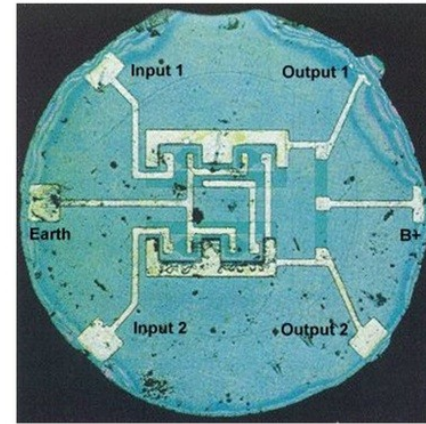
- Transistor, Resistors and Capacitors on the same piece of semiconductor
- **Interconnects between components not integrated**  
→ Low connectivity between components



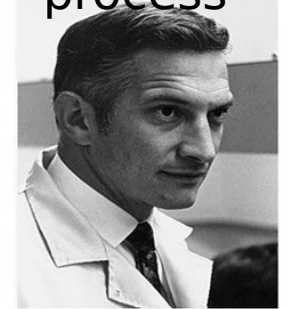
## The First **Monolithic** (2D) Integrated Circuit

Robert Noyce, Fairchild Semiconductor, 1961

- Transistor, Resistors and Capacitors on the same piece of semiconductor
- **Interconnects between components integrated**  
→ High connectivity between components



Used  
planar  
process



**Jean Hoerni's** development of the **planar process** in 1959 enabled the fabrication of transistors and other components on a flat silicon wafer. This breakthrough was key to integrating multiple components on a single chip. The process uses **photolithography** to etch intricate patterns onto silicon, enabling mass production of precise and consistent circuits.



# How is a Semiconductor Device Manufactured?

Wafer  
Fabricatio  
n

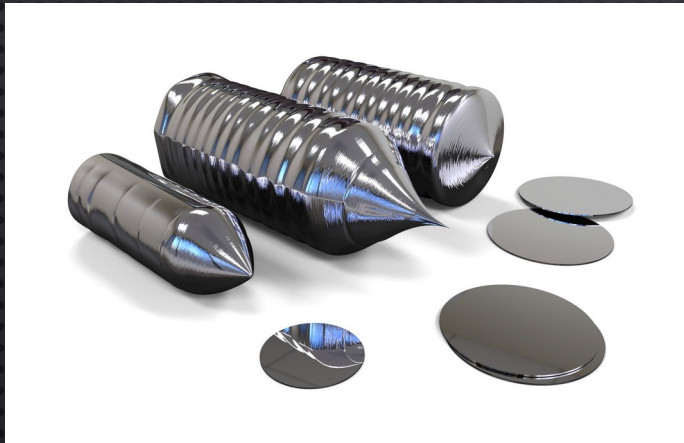
Photoresi  
st Coating

Photo-  
Lithograp  
hy

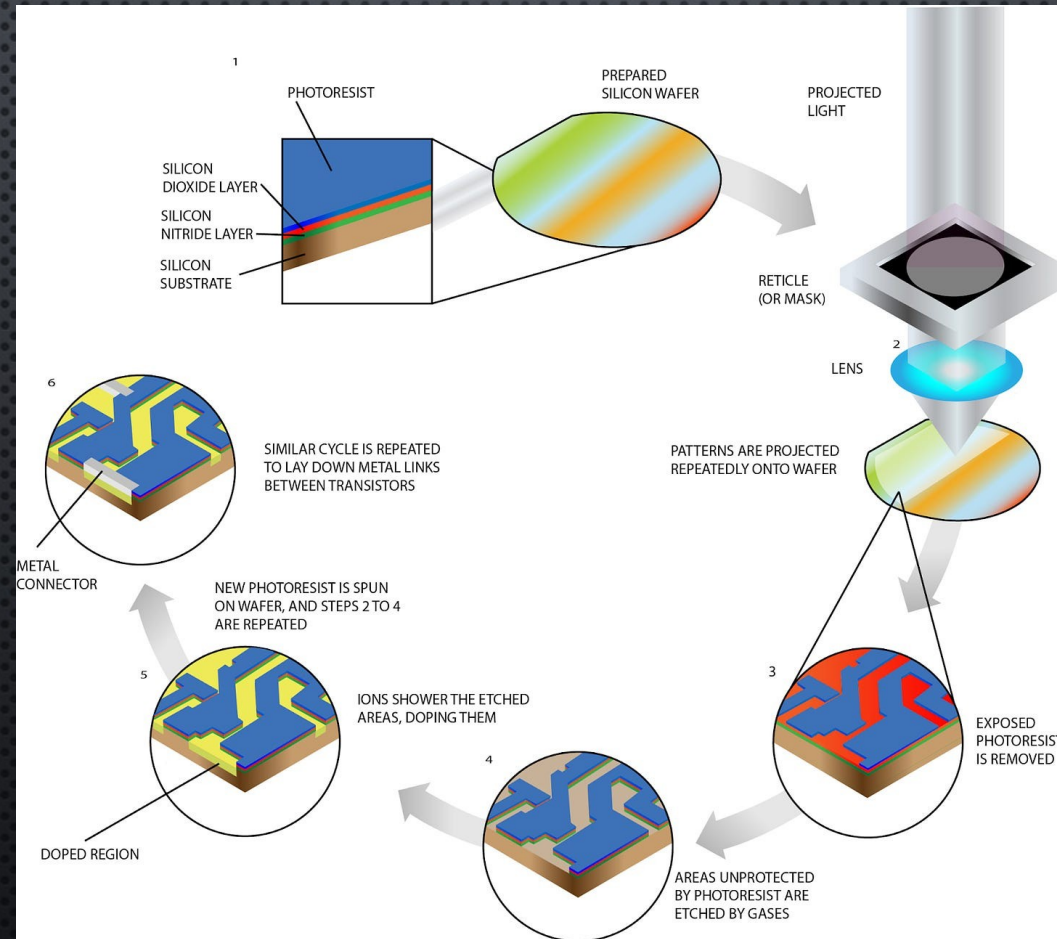
Etching

Ion  
Implantati  
on

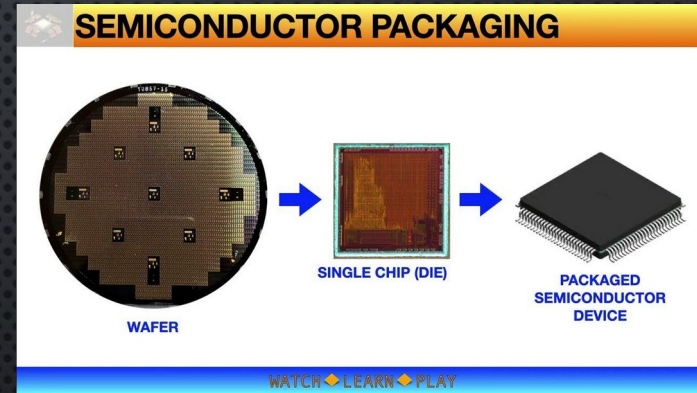
Assembly  
&  
Packaging



<https://www.waferworld.com/>



<https://www.generativevalue.com/p/an-overview-of-the-semiconductor>

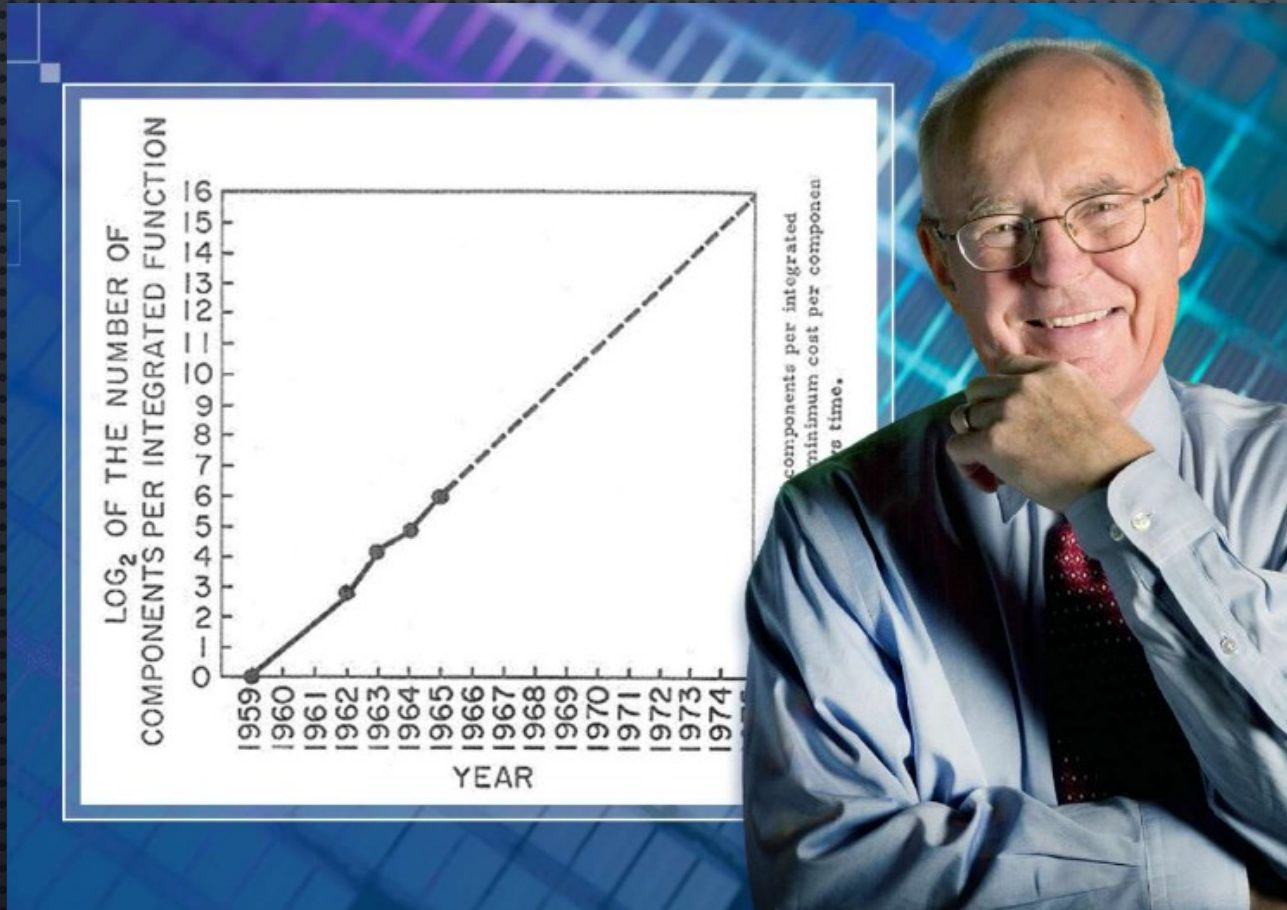


<https://www.youtube.com/watch?v=DiFovfCtvgw>



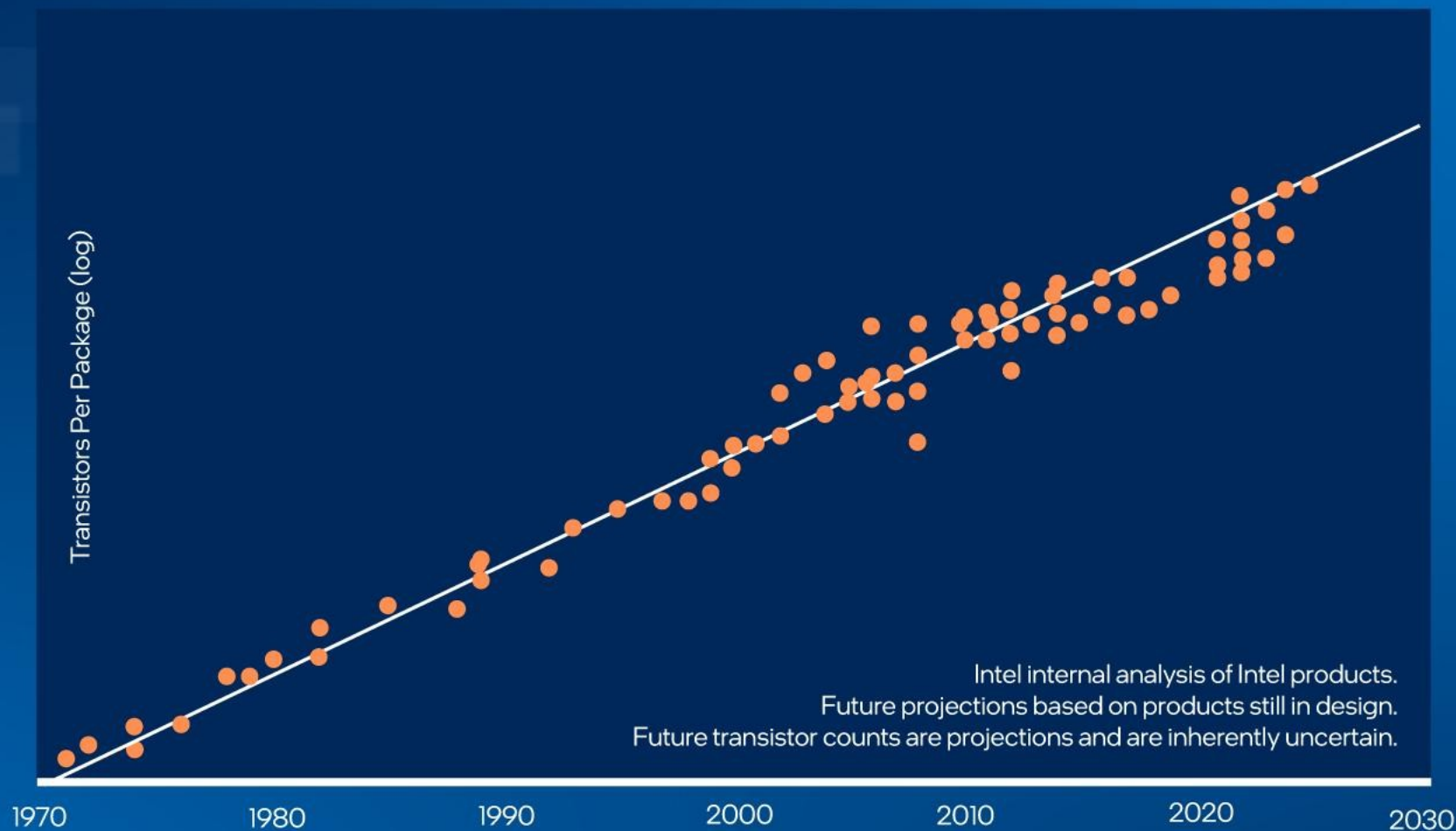
# Along comes Gordon Moore...

Gordon Moore hypothesized that the **number of transistors on an integrated circuit will double every two years** with minimal rise in cost.



“Over the years, many have predicted the death of Moore’s Law – an inevitable end to the innovation due to the constraints of present-day methods: You can make something smaller and smaller ... until you can’t. Then, you must innovate in other ways. With advanced packaging technology and new materials, the pace of Moore’s Law continues.”



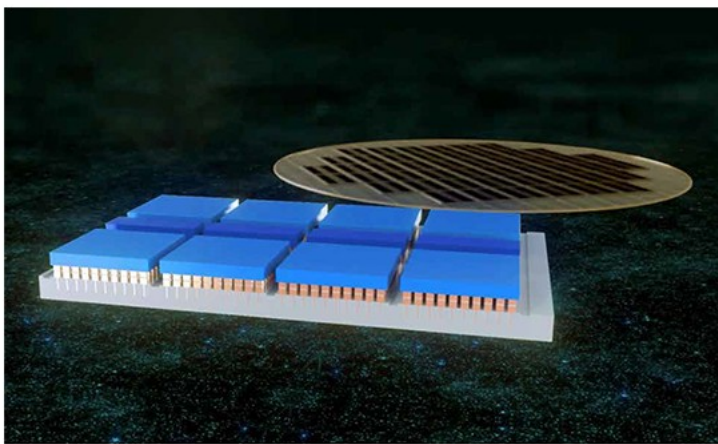


Aspiring to  
**1 Trillion**  
transistors in 2030

- ✓ RibbonFET
- ✓ PowerVia
- ✓ High NA
- ✓ 2.5D/3D packaging

In December 2022, on the 75th anniversary of the transistor, Intel announced that it had identified new materials and processes that blur the line between packaging and silicon and revealed critical next steps on the journey to extending Moore's Law to 1 trillion transistors on a package

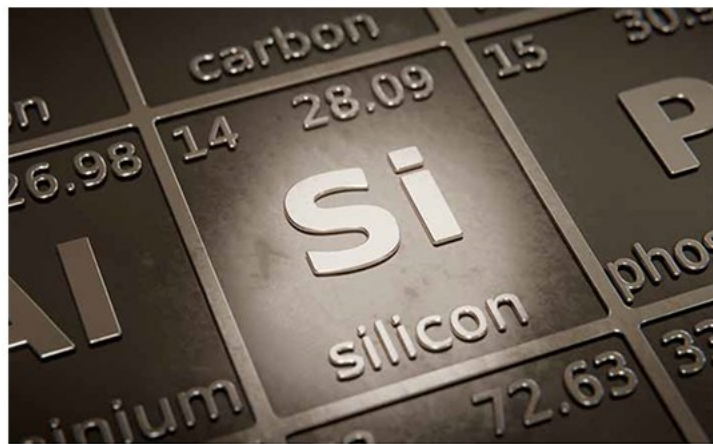




## Delivering More Transistors

Essential scaling technologies for delivering more transistors include making them faster and smaller, so we can deliver millions more per square area and deliver them as tiles, or chiplets, via advanced packaging.

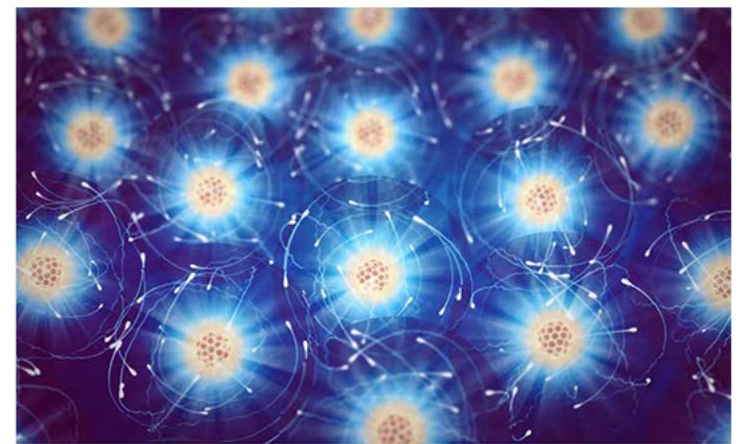
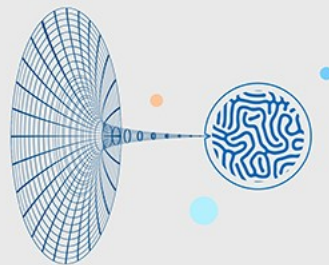
- Interconnect density improvement in packaging
- Area improvement with 3D transistor stacking
- Super thin materials for future scaling



## Bringing New Capabilities to Silicon

As we enable more powerful computing through scaling, we need to stretch the limits of silicon and integrate new materials so we can deliver power more efficiently and meet greater demands for memory.

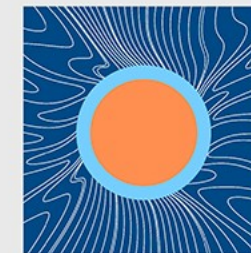
- World's first integration of GaN-based power switches
- Record FeRAM speed and endurance



## Embracing the Quantum Realm

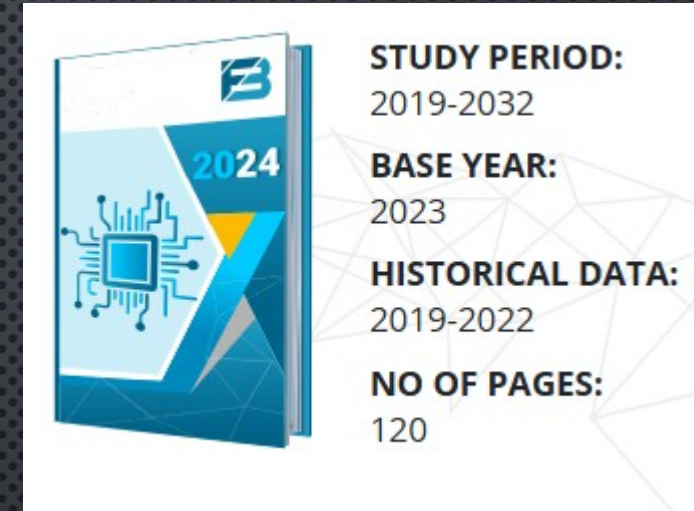
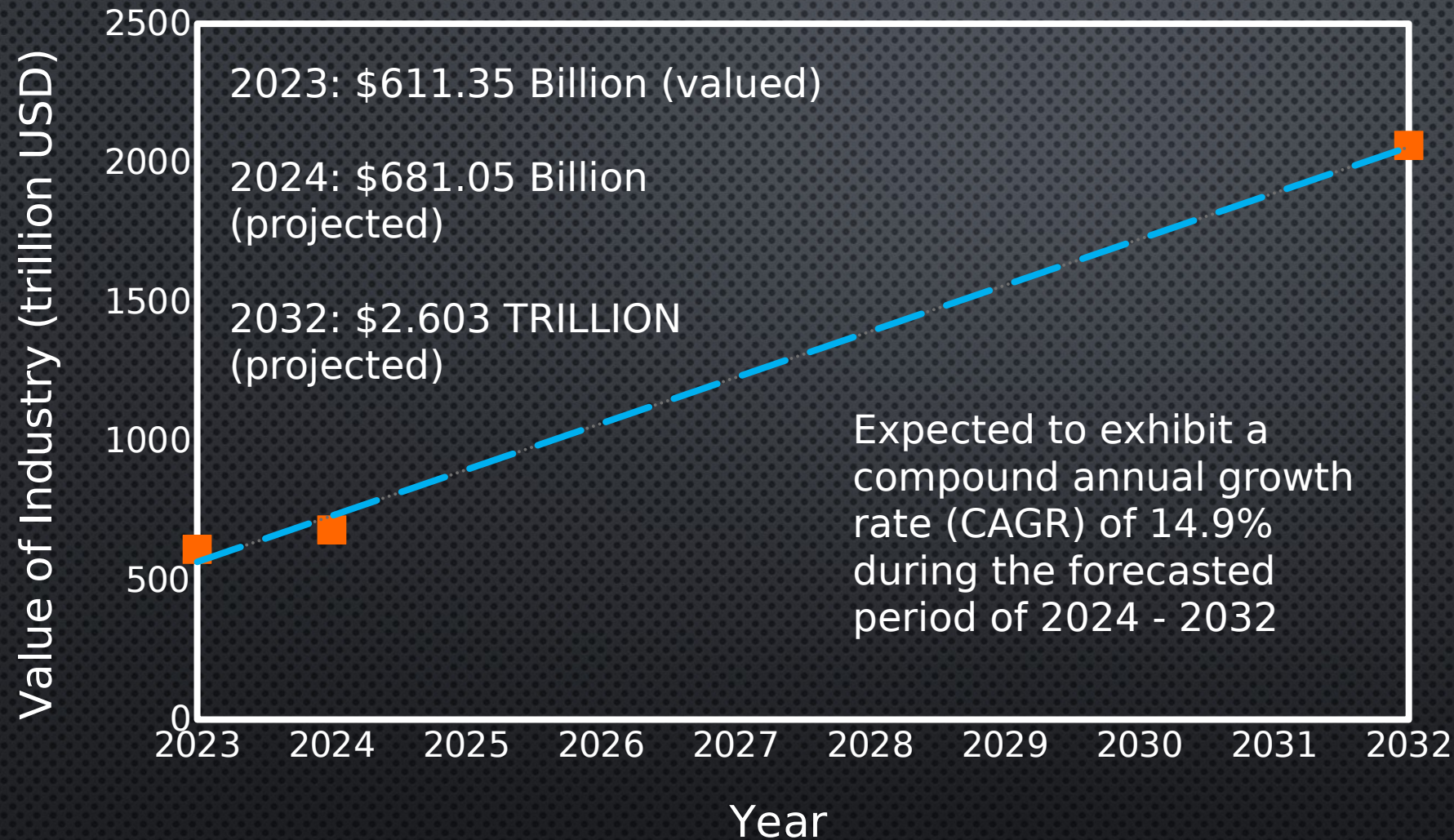
We are exploring entirely new concepts in physics that may one day revolutionize computing by potentially replacing classic transistors, enabling even greater performance and power efficiencies.

- Magnetoelectric spin-orbit logic
- Spin-torque devices
- 300mm qubit process flows





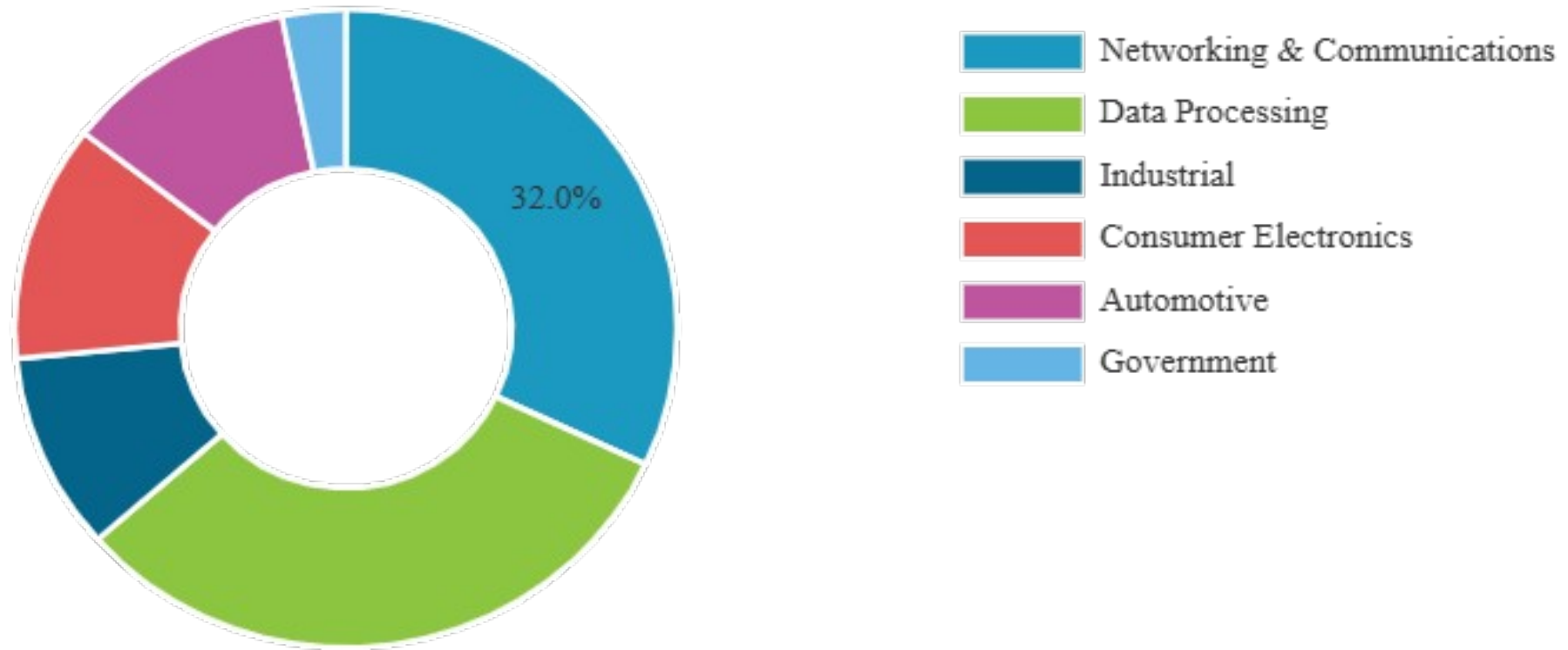
# The Semiconductor Industry: Global Market Size





# The Semiconductor Industry: Market Segmentation

Global Semiconductor Market Share, By Application, 2023



[www.fortunebusinessinsights.com](http://www.fortunebusinessinsights.com)



# The Semiconductor Industry

The semiconductor industry has two main business models: either the company designs and manufactures its own chips, or it designs them and then outsources their manufacture.

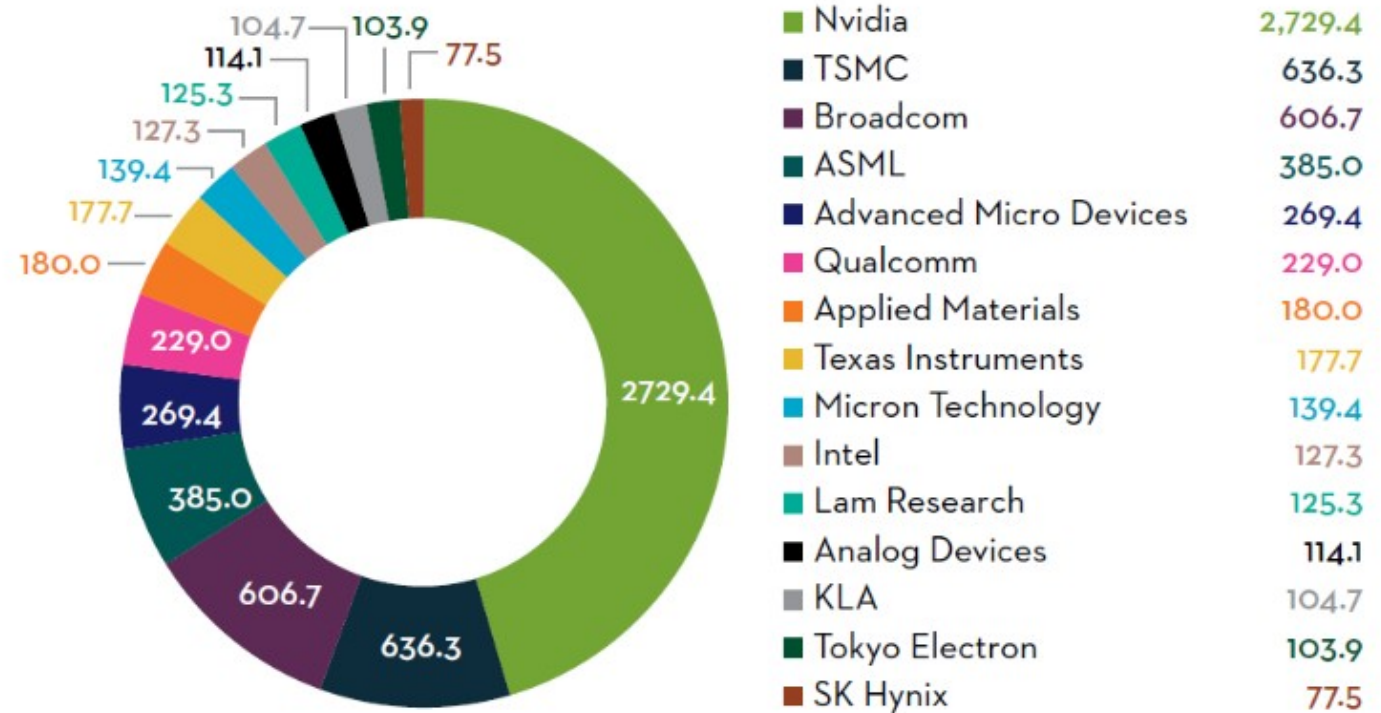
Companies operating the former operating model are called **integrated device manufacturers** (IDMs).

Those operating the latter model are called **fabless** semiconductor companies, outsourcing the manufacturing of their designs to foundries.

The majority of companies rely heavily on **foundries** for their chips.

<https://www.martincurrie.com/insights/a-beginners-guide-to-semiconductors>

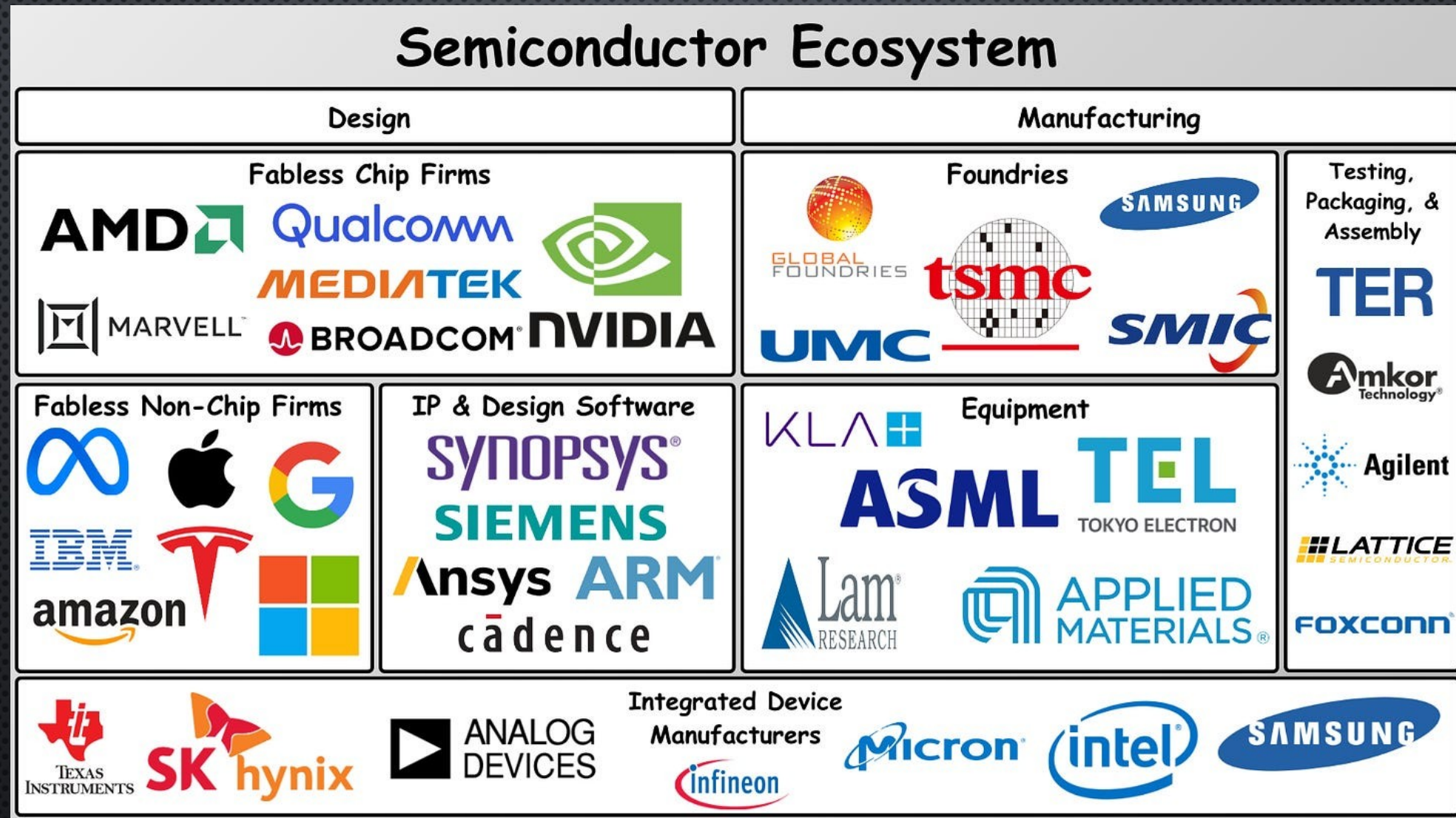
Top 15 semiconductor companies worldwide market capitalisation  
(at December 2022 in billion US dollars)



Source: FactSet at 30 May 2024. Using market capitalisation denoted in billion US dollars, based on publicly traded companies and Factset industry classification of these companies as Semiconductor companies with the following definition "This industry group consists of companies engaged in the manufacturing of integrated circuits for electronic applications."



# The Semiconductor Industry: Key Players





# The Semiconductor Industry: Job Creation

THE WHITE HOUSE



AUGUST 09, 2022

## FACT SHEET: CHIPS and Science Act Will Lower Costs, Create Jobs, Strengthen Supply Chains, and Counter China

The **CHIPS and Science Act** is a U.S. federal statute that was signed into law by President Joe Biden on August 9, 2022.  
**A \$52.7 Billion Investment**

Alongside the enactment of this act, more than \$50 billion dollars in additional investments in the American semiconductor industry were announced:



\$40 Billion investment in memory chip manufacturing that will create 40,000 new US jobs in construction and manufacturing, bringing US market share from less than 2% up to 10% over next decade



Announced a new partnership that includes a \$4.2 Billion investment in GlobalFoundries' upstate New York facility



<https://www.whitehouse.gov/briefing-room/statements-releases/2022/08/09/fact-sheet-chips-and-science-act-will-lower-costs-create-jobs-strengthen-supply-chains-and-counter-china/>



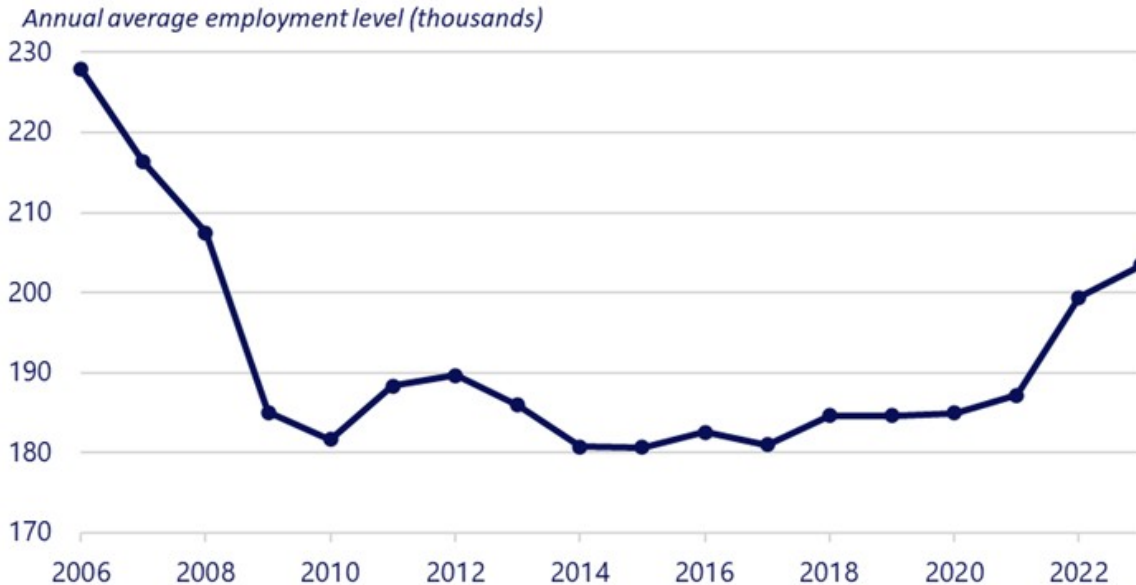


# The Semiconductor Industry: Job Creation

MARCH 20, 2024

## U.S. Semiconductor Jobs are Making a Comeback

Figure 1: Employment in Semiconductor Manufacturing



Council of Economic Advisers

Source: Bureau of Labor Statistics.

Note: Graph shows employment in NAICS 334413, Semiconductor and Related Device Manufacturing. 2023 annual average includes data through September.

As of March 6, 2024 at 4:00pm.

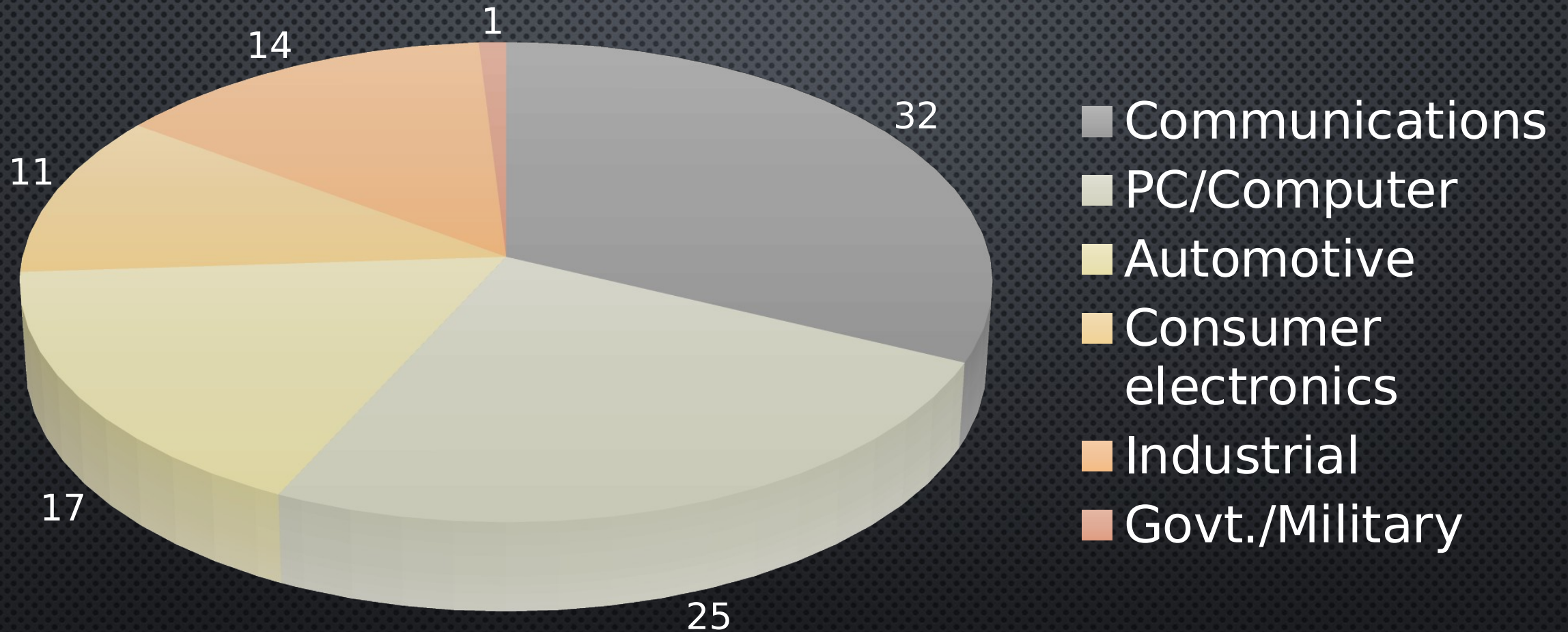
“America invented the semiconductor. But in the 2000s and early 2010s, many semiconductor firms closed U.S. plants and shifted production to lower-cost locals overseas.”

“Hiring in the U.S. surged in 2021 and 2022 during the global chips shortage, as legacy semiconductor producers in the U.S. ramped up hiring at existing U.S. Facilities.”

“Since 2021, semiconductor employment has increased an average of 4.3% per year to over 203,000 workers in 2023. Between 2022:Q2 and 2023:Q3, the most recent period of available data, semiconductor manufacturers added about 6,000 jobs.”



# The Semiconductor Industry: End-Market Industrial Usage (in %)



Source: World Semiconductor Trade Statistics



# The Semiconductor Industry: Cultural Impacts

Semiconductors have been fundamental to the development of both the hardware and the communication technologies that underpin the internet.



Without the advances in semiconductor technology, the internet as we know it today—ubiquitous, fast, and accessible—would not have been possible.

Image taken from: <https://mbrjournal.com/>



# Challenges facing the Semiconductor Industry.

The semiconductor industry faces several critical challenges as it evolves to meet growing demand and technological advancements. Here are the most prominent challenges:

1. Global supply chain issues
2. Rising costs of advanced manufacturing
3. Geopolitics
4. Talent shortages
5. Technological barriers



# Challenges: Global Supply Chain Issues

The industry heavily depends on a few regions (like Taiwan, South Korea, and China) for semiconductor manufacturing, leading to vulnerabilities in the supply chain.

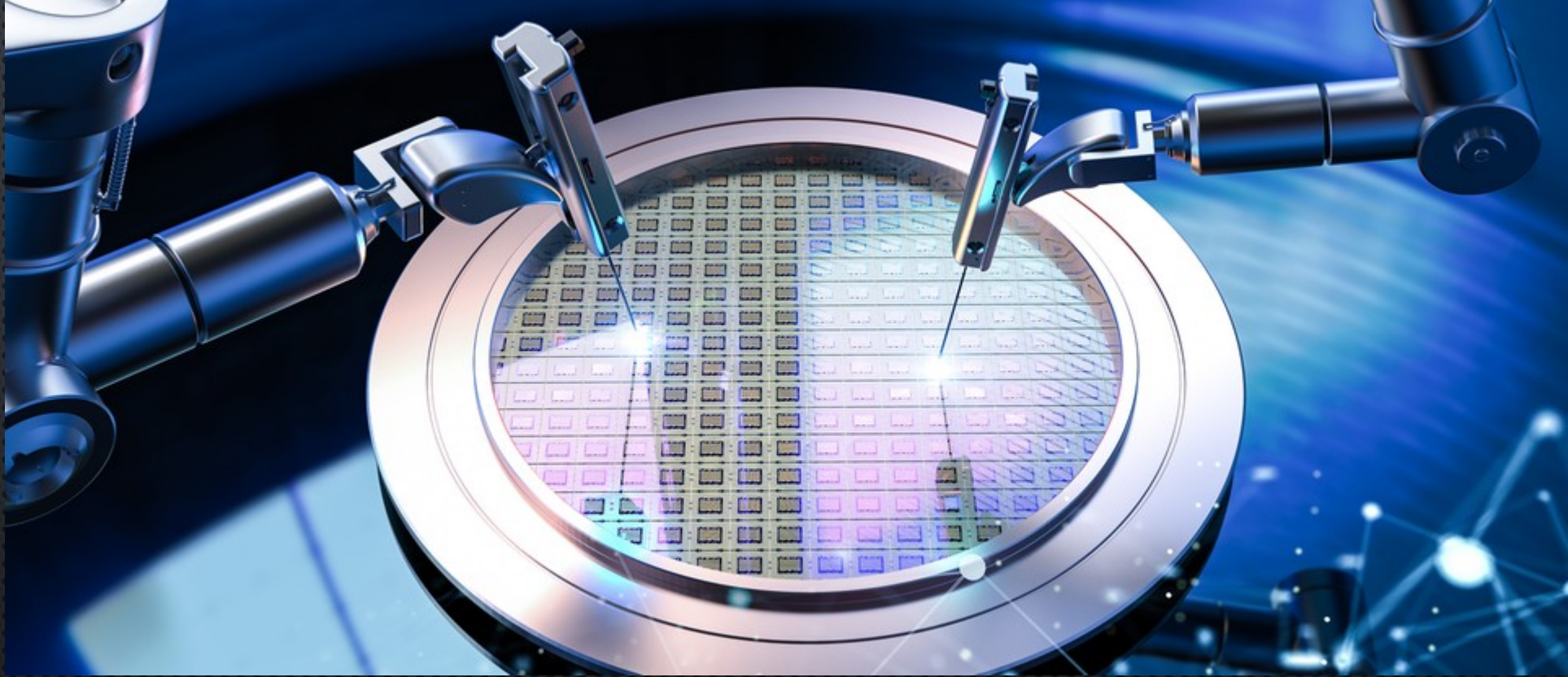


Rising costs of raw materials, shipping delays, and supply chain disruptions have increased production costs and lead times.



# Challenges: Rising Cost of Advanced Manufacturing

Developing next-generation semiconductor technology (e.g., 3nm and smaller nodes) requires substantial investment in research and development.



Building state-of-the-art fabrication plants (fabs) costs billions of dollars. This high cost of entry into the market limits competition and innovation from smaller players.



# Challenges: Geopolitics

Export controls and restrictions on semiconductor equipment and materials between China and the United States create uncertainties.



Taiwan, home to TSMC (the world's largest chipmaker), resides geopolitically-sensitive region, creating risks for global semiconductor supply.



# Challenges: Talent Shortages

The industry is experiencing a lack of skilled engineers and technicians needed for semiconductor design and manufacturing.



As more industries become tech-driven (e.g., automotive, AI), competition for semiconductor experts has intensified.



# Challenges: Technological barriers

Shrinking transistors further is becoming increasingly difficult due to physical and quantum limitations, making it hard to keep up with Moore's law, which has driven the industry for some time.



As chips become more powerful, improving energy efficiency to prevent overheating and reduce power consumption is a major hurdle. Also, developing chips tailored for diverse applications, such as AI, 5G, IoT, and the automotive industry, requires specialized innovations.



# Where it all started...

## 1833

Michael Faraday discovered that the electrical conductivity of silver sulfide crystal increases with temperature.

## 1947

A team of three American scientists, John Bardeen, Walter Houser Brattain and William Shockley realised semiconductors made of germanium could actively influence the flow of current. The 1947 discovery spurred the birth of Silicon Valley.

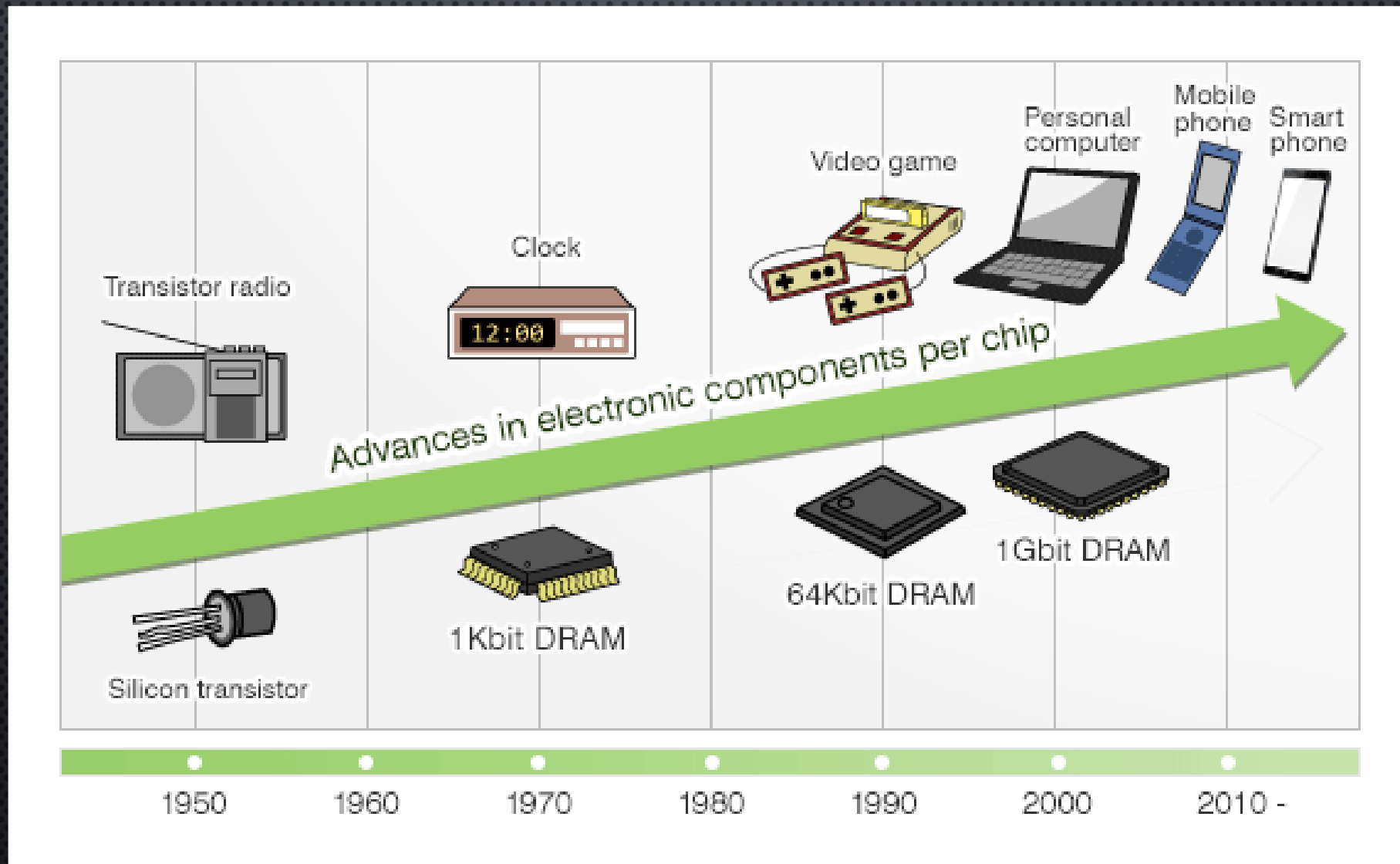
## 1874

Karl Braun discovered and documented the first semiconductor diode effect.

Timeline of key events in the history of semiconductors



# What it triggered...





# Where will we end up...



## Future Trends Driving the Semiconductor Industry





# QUESTIONS

# ?